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GENERALIZABLE END-TO-END AUTONOMOUS DRIVING WITH MULTI-MODAL FOUNDATION MODELS

Abstract

Systems and methods described herein relate to using multimodal foundation models. In one embodiment, a method includes receiving images and a foundation multi-model, selecting a mask set, modifying the foundation multi-model to include query, key, and value matrices, and applying the mask set to the foundation multi-model to obtain patch-aligned features.

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Background/Summary

TECHNICAL FIELD

[0001] The subject matter described herein relates, in general, to strategies for using multi-modal foundation models as a basis for an end-to-end driving model.

BACKGROUND

[0002] In the pursuit of autonomous driving, an end-to-end methodology may provide a holistic construction of the system that encompasses a large variety of issues from perception to control. However, prevalent systems may exhibit the following prominent limitations: [0003] (i) Open set environments: self-driving vehicles may operate in extremely diverse scenarios that are impractical to fully capture within training datasets. When these systems encounter situations that deviate from what they've learned (e.g., out-of-distribution (OOD) data), performance may deteriorate, giving rise to uncertainty and potential safety risks. [0004] (ii) Black-/gray-box models: the use of complex, advanced machine learning models may complicate the task of pinpointing the root causes of failures in autonomous systems. For example, identifying which learned concepts, objects, or even individual pixels contributed to incorrect behavior may be a difficult task.

SUMMARY

[0005] In one embodiment, a system is disclosed. The vehicle management system includes one or more processors and a memory communicably coupled to the one or more processors. The memory stores a command module including instructions that when executed by the one or more processors cause the one or more processors to receive images and a foundation multi-model, select a mask set, modify the foundation multi-model to include query, key, and value matrices, and apply the mask set to the foundation multi-model to obtain patch-aligned features.

[0006] In one embodiment, a non-transitory computer-readable medium including instructions that when executed by one or more processors cause the one or more processors to perform one or more functions is disclosed. The instructions include instructions to receive images and a foundation multi-model, select a mask set, modify the foundation multi-model to include query, key, and value matrices, and apply the mask set to the foundation multi-model to obtain patch-aligned features.

[0007] In one embodiment, a method is disclosed. In one embodiment, the method includes receiving images and a foundation multi-model, selecting a mask set, modifying the foundation multi-model to include query, key, and value matrices, and applying the mask set to the foundation multi-model to obtain patch-aligned features.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

[0009] FIG. 1 illustrates one embodiment of a vehicle within which systems and methods disclosed herein may be implemented.

[0010] FIG. 2 illustrates one embodiment of an autonomous driving system that is associated with multimodal foundation models.

[0011] FIG. **3** illustrates one embodiment of a cloud computing environment within which the systems and methods described herein may operate.

[0012] FIG. **4** illustrates one example of patch-wise feature extraction.

[0013] FIG. **5** illustrates one example of language-augmented latent space simulation.

[0014] FIG. **6** illustrates one example of a method for using multimodal foundation models.

DETAILED DESCRIPTION

[0015] Systems, methods, and other embodiments associated with multi-modal foundation models. Traditional end-to-end models typically cannot handle situations involving OOD data all that well. In comparison, an end-to-end model is described herein that utilizes attention masks and matrixes in relation to a multimodal foundation to modal to characterize a patch by both its location and language features. Further, human priors or large language models may be used to determine associated concepts similar to the patch-aligned features. For example, a patch-aligned feature for trees in a rural context may also be similar to a lamp post textual feature in an urban context. Accordingly, by replacing patch-aligned features with textual features based on a shared similarity between them, the end-to-end model may be contextually adjusted to operate within an OOD environment.

[0016] Referring to FIG. **1**, an example of a vehicle **100** is illustrated. As used herein, a “vehicle” is any form of motorized transport. In one or more implementations, vehicle **100** is an automobile. While arrangements will be described herein with respect to automobiles, it will be understood that embodiments are not limited to automobiles. In some implementations, vehicle **100** may be any robotic device or form of motorized transport that, for example, includes sensors to perceive aspects of the surrounding environment, and thus benefits from the functionality discussed herein associated with strategies for using multi-modal foundation models. As a further note, this disclosure generally discusses vehicle **100** as traveling on a roadway with surrounding vehicles, which are intended to be construed in a similar manner as vehicle **100** itself. That is, the surrounding vehicles may include any vehicle that may be encountered on a roadway by vehicle **100**.

[0017] Vehicle **100** also includes various elements. It will be understood that in various embodiments it may not be necessary for vehicle **100** to have all of the elements shown in FIG. **1**. Vehicle **100** may have any combination of the various elements shown in FIG. **1**. Further, vehicle **100** may have additional elements to those shown in FIG. **1**. In some arrangements, vehicle **100** may be implemented without one or more of the elements shown in FIG. **1**. While the various elements are shown as being located within vehicle **100** in FIG. **1**, it will be understood that one or more of these elements may be located external to vehicle **100**. Further, the elements shown may be physically separated by large distances. For example, as discussed, one or more components of the disclosed system may be implemented within a vehicle while further components of the system are implemented within a cloud-computing environment or other system that is remote from vehicle **100**.

[0018] Some of the possible elements of vehicle **100** are shown in FIG. **1** and will be described along with subsequent figures. However, a description of many of the elements in FIG. **1** will be provided after the discussion of FIGS. **2-6** for purposes of brevity of this description. Additionally, it will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, the discussion outlines numerous specific details to provide a thorough understanding of the embodiments described herein. Those of skill in the art, however, will understand that the embodiments described herein may be practiced using various combinations of these elements. In either case, vehicle **100** includes an E2E driving system **170** that is implemented to perform methods and other functions as disclosed herein. As will be discussed in greater detail subsequently, E2E driving system **170**, in various embodiments, is implemented partially within vehicle **100** and as a cloud-based service. For example, in one approach, functionality associated

with at least one module of E2E driving system **170** is implemented within vehicle **100** while further functionality is implemented within a cloud-based computing system.

[0019] With reference to FIG. **2**, one embodiment of E2E driving system **170** of FIG. **1** is further illustrated. E2E driving system **170** is shown as including processor(s) **110** from vehicle **100** of FIG. **1**. Accordingly, processor(s) **110** may be a part of E2E driving system **170**, E2E driving system **170** may include a separate processor from processor **110** (s) of vehicle **100**, or E2E driving system **170** may access processor **110** (s) through a data bus or another communication path. In one embodiment, E2E driving system **170** includes memory **210**, which stores detection module **220** and command module **230**. Memory **210** is a random-access memory (RAM), read-only memory (ROM), a hard-disk drive, a flash memory, or other suitable memory for storing detection module **220** and command module **230**. Detection module **220** and command module **230** are, for example, computer-readable instructions that when executed by processor(s) **110** cause processor(s) **110** to perform the various functions disclosed herein.

[0020] E2E driving system **170** as illustrated in FIG. **2** is generally an abstracted form of E2E driving system **170** as may be implemented between vehicle **100** and a cloud-computing environment. Accordingly, E2E driving system **170** may be embodied at least in part within a cloud-computing environment to perform the methods described herein.

[0021] With reference to FIG. **2**, detection module **220** generally includes instructions that function to control processor(s) **110** to receive data inputs from one or more sensors of vehicle **100**. The inputs are, in one embodiment, observations of one or more objects in an environment proximate to vehicle **100**, other aspects about the surroundings, or both. As provided for herein, detection module **220**, in one embodiment, acquires sensor data **250** that includes at least camera images. In further arrangements, detection module **220** acquires sensor data **250** from further sensors such as radar **123**, LiDAR **124**, and other sensors as may be suitable for identifying vehicles, locations of the vehicles, lane markers, crosswalks, traffic signs, vehicle parking areas, road surface types, curbs, vehicle barriers, and so on.

[0022] Accordingly, detection module **220**, in one embodiment, controls the respective sensors to provide sensor data **250**. Additionally, while detection module **220** is discussed as controlling the various sensors to provide sensor data **250**, in one or more embodiments, detection module **220** may employ other techniques to acquire sensor data **250** that are either active or passive. For example, detection module **220** may passively sniff sensor data **250** from a stream of electronic information provided by the various sensors to further components within vehicle **100**. Moreover, detection module **220** may undertake various approaches to fuse data from multiple sensors when providing sensor data **250**, from sensor data acquired over a wireless communication link (e.g., v2v) from one or more of the surrounding vehicles, or from a combination thereof. Thus, sensor data **250**, in one embodiment, represents a combination of perceptions acquired from multiple sensors.

[0023] In addition to locations of surrounding vehicles, sensor data **250** may also include, for example, odometry information, GPS data, or other location data. Moreover, detection module **220**, in one embodiment, controls the sensors to acquire sensor data about an area that encompasses 360 degrees about vehicle **100**, which may then be stored in sensor data **250**. In some embodiments, such area sensor data may be used to provide a comprehensive assessment of the surrounding environment around vehicle **100**. Of course, in alternative embodiments, detection module **220** may acquire the sensor data about a forward direction alone when, for example, vehicle **100** is not equipped with further sensors to include additional regions about the vehicle or the additional regions are not scanned due to other reasons (e.g., unnecessary due to known current conditions).

[0024] Moreover, in one embodiment, E2E driving system **170** includes a database **240**. Database **240** is, in one embodiment, an electronic data structure stored in memory **210** or another data store and that is configured with routines that may be executed by processor(s) **110** for analyzing stored data, providing stored data, organizing stored data, and so on. Thus, in one embodiment, database

240 stores data used by the detection module **220** and command module **230** in executing various functions. In one embodiment, database **240** includes sensor data **250** along with, for example, metadata that characterize various aspects of sensor data **250**. For example, the metadata may include location coordinates (e.g., longitude and latitude), relative map coordinates or tile identifiers, time/date stamps from when separate sensor data **250** was generated, and so on. [0025] In one embodiment, command module **230** generally includes instructions that function to control the processor(s) **110** or collection of processors in the cloud-computing environment **300** as shown in FIG. 3.

[0026] With reference to FIG. 3, vehicle **100** may be connected to a network **305**, which allows for communication between vehicle **100** and cloud servers (e.g., cloud server **310**), infrastructure devices (e.g., infrastructure device **340**), other vehicles (e.g., vehicle **380**), and any other systems connected to network **305**. With respect to network **305**, such a network may use any form of communication or networking to exchange data, including but not limited to the Internet, Directed Short Range Communication (DSRC) service, LTE, 5G, millimeter wave (mmWave) communications, and so on.

[0027] Cloud server **310** is shown as including a processor **315** that may be a part of E2E driving system **170** through network **305** via communication unit **335**. In one embodiment, cloud server **310** includes a memory **320** that stores a communication module **325**. Memory **320** is a random-access memory (RAM), read-only memory (ROM), a hard-disk drive, a flash memory, or other suitable memory for storing communication module **325**. Communication module **325** is, for example, computer-readable instructions that when executed by processor **315** causes processor **315** to perform the various functions disclosed herein. Moreover, in one embodiment, cloud server **310** includes database **330**. Database **330** is, in one embodiment, an electronic data structure stored in a memory **320** or another data store and that is configured with routines that may be executed by processor **315** for analyzing stored data, providing stored data, organizing stored data, and so on.

[0028] Infrastructure device **340** is shown as including a processor **345** that may be a part of E2E driving system **170** through network **305** via communication unit **370**. In one embodiment, infrastructure device **340** includes a memory **350** that stores a communication module **355**. Memory **350** is a random-access memory (RAM), read-only memory (ROM), a hard-disk drive, a flash memory, or other suitable memory for storing communication module **355**. Communication module **355** is, for example, computer-readable instructions that when executed by processor **345** causes processor **345** to perform the various functions disclosed herein. Moreover, in one embodiment, infrastructure device **340** includes a database **360**. Database **360** is, in one embodiment, an electronic data structure stored in memory **350** or another data store and that is configured with routines that may be executed by processor **345** for analyzing stored data, providing stored data, organizing stored data, and so on.

[0029] Accordingly, in addition to information obtained from sensor data **250**, E2E driving system **170** may obtain information from cloud servers (e.g., cloud server **310**), infrastructure devices (e.g., infrastructure device **340**), other vehicles (e.g., vehicle **380**), and any other systems connected to network **305**. For example, cloud servers (e.g., cloud server **310**) may be used to perform the same tasks as described herein with respect to command module **230**.

[0030] In some embodiments, command module **230** may include a control system ϕ providing steering and acceleration commands based on a stream of perception data $F \in \mathbb{R}^{\text{custom-character} \times \text{sup.H} \times \text{W} \times 3}$ obtained via sensor data **250** (e.g., RGB images acquired through vehicle-mounted sensors $u = \phi(F)$). The control system may be further enhanced by substituting in place of the perception data F a dense feature representation $F' \in \mathbb{R}^{\text{custom-character} \times \text{sup.H}' \times \text{W}' \times D}$ extracted via a multimodal foundation model Desc, where (H', W') may denote the resolution of the dense features in the spatial dimensions and D is the number of channels. With respect to Desc, command module **230** may employ a multimodal foundation model such as Contrastive Language-Image Pre-Training (“CLIP”), self DIsillation, NO labels (“DINO”), Bootstrapping Language-Image Pre-

Training (“BLIP”), or other suitable foundation models. Any of these multimodal foundation models or other models may be stored in predictive model **260** for use by command module **230**. In addition, the control system Φ may be implemented for example by control barrier functions stored in predictive model **260**.

[0031] Accordingly, once a multimodal foundation model Desc:  custom-character .sup.H×W×3.fwdarw. custom-character.sup.D of L layers, an input image/frame $F \in \text{img_custom-character.sup.H} \times \text{img_custom-character.sup.W} \times 3$, and a desired resolution $H' \leq H$ and $W' \leq W$ are selected, command module **230** may extract a feature descriptors tensor $F' \in \text{img_custom-character.sup.H'} \times \text{img_custom-character.sup.W'} \times D$. For example, when applying Desc on F where $N=H'W'$, for every layer $l \in [L]$ (where [i] denotes the set $\{1, \dots, i\}$), command module **230** may use $Q.\text{sub.Desc}(F).\text{sup.l} \in \text{img_custom-character.sup.N} \times D.\text{sup.k}$ and $V.\text{sub.Desc}(F).\text{sup.l} \in \text{img_custom-character.sup.N} \times D$ to denote respectively the resulted query, key, and value matrices in the $l.\text{sup.th}$ attention layer. Next, so as to extract features for a specific patch (or area in the image) $F'.\text{sup.(j)}$, command module **230** may use an attention mask $m.\text{sup.(j)} = (m.\text{sub.1.sup.(j)}, \dots, m.\text{sub.2.sup.(j)}) \in \text{img_custom-character.sup.N}$, where each element $m.\text{sub.i.sup.(j)} \in [0,1]$ may determine how much the i th patch of vector $m.\text{sup.(j)}$ contributes to the desired patch feature $F'.\text{sup.(j)}$, as follows:

[0032] First, command module **230** may select a value of r , where $r \in (-\infty, 0)$, to control the strength of the masking (e.g., the larger the value of $|r|$, the higher the effects of masking).

[0033] Next, command module **230** may define the matrix $G.\text{sub.Desc}(F).\text{sup.l}$ as the matrix multiplication of the key and query matrices at the l th attention layer:

$$[00001] \quad G_{Desc(F)}^l := Q_{Desc(F)}^l (K_{Desc(F)}^l)^T \quad \text{Equation(1)}$$

[0034] Given a matrix $M.\text{sup.(j)} = [m.\text{sup.(j)}, \dots, m.\text{sup.(j)}].\text{sup.T} \in \text{img_custom-character.sup.N} \times N$, command module **230** may then obtain a masked version of $G.\text{sub.Desc}(F).\text{sup.l}$ as follows:

$$[00002] \quad \hat{G}_{Desc(F)}^{l,(j)} := G_{Desc(F)}^l + (1 - M^j) \cdot \text{Math. } r, \quad \text{Equation(2)}$$

where $1 \in \text{img_custom-character.sup.N} \times N$ is an all-ones matrix.

[0035] Equation (2) may be interpreted as setting the attention scores (in the matrix $\hat{G}.\text{sub.Desc}(F).\text{sup.l,(j)}$) for “non-contributing” patches (e.g., where their corresponding $m.\text{sub.i}$ is close to 0) to be close to r (e.g., low value), thereby masking them out. In addition, the term $(1 - M.\text{sup.j})$ may ensure that the patches with a corresponding attention mask equal to 1 have an added softmax'ed score of 0 (e.g., no modification), and a very low value (e.g., effectively r) if the corresponding attention mask is near 0.

[0036] After obtaining the modified attention scores, command module **230** may obtain the final attention weights with a softmax function as follows:

$$[00003] \quad F^{(j)} := Desc^l.fwdarw. (\text{SoftMax}(\hat{G}_{Desc(F)}^{l,(j)}) (V_{Desc(F)}^l)^T), \quad \text{Equation(3)}$$

where $Desc.\text{sup.l.fwdarw.}$ is the rest of the foundation model after the l th layer.

[0037] In some embodiments, the approach described above can be extended to any region-wise feature extraction by generalizing the definition of patches to arbitrarily-shaped regions.

[0038] In some embodiments, the i th entry of $m.\text{sup.j}$ may be defined to correspond to an expectation of how much patch i contributes to the semantic information of patch j . For example, “close” neighbors may be expected to contribute more than ones further away. As such, if $(x.\text{sub.i}, y.\text{sub.i})$ is the row-stacked ordering of the image grid after patching, then $\text{dist}(i, j)$ may denote the distance between patch i and patch j as follows:

$$[00004] \quad \text{dist}(i, j) := \text{Math. } (x_i, y_i) - (x_j, y_j) \cdot \text{Math. } \frac{1}{z}, \quad \text{Equation(4)}$$

where $z \geq 1$ may define the norm. And based on Equation (4), a value for $m.\text{sub.i.sup.(j)}$ may be obtained as follows:

$$[00005] \quad m_i^{(j)} := f(\text{dist}(i, j)), \quad \text{Equation(5)}$$

where f may be defined as

$$[00006] \begin{cases} 0, & \text{if } \text{dist}(0, 1) > \\ 1, & \text{otherwise} \end{cases}, \quad 1 / 2^{\text{dist}(i,j)}, \quad 1 / \text{dist}(i,j),$$

or other functions.

[0039] In some embodiments, increasing the spatial resolution may enhance the foundational model's spatial features, as the higher resolution may allow for more granular, non-overlapping patches. In some embodiments, a pretrained Vision Transformer (“ViT”) may be used to extract overlapping patches during inference, so as to interpolate their position encoding. In this manner, the approach may yield multimodal features with a finer spatial resolution, which may also be obtained without requiring additional training.

[0040] With respect to FIG. 4, a representation of a method for patch-wise feature extraction that may preserve spatial information critical for end-to-end driving is shown based on the approach described herein, which may involve constructing attention masks anchored at each patch location to focus on specific regions for the attention module.

[0041] As each patch feature $F'.\text{sup.}(j)$ may incorporate language modality, the approach described herein may be further integrated with one or more large language models (“LLMs”) as shown in FIG. 5. Such an integration may allow for conducting latent space simulations, where $F'.\text{sup.}(j)$ is replaced with alternate textual features to simulate different scenarios. For example, command module 230 may receive a set of concepts in natural language (e.g., relevant to autonomous driving) from LLMs and compute their corresponding textual feature as follows:

$$[00007] T_k = \text{Desc}(c_k), \quad \text{Equation(6)}$$

where $c.\text{sub.}k \in C.\text{sup.}.\text{src}/\text{tgt} = \text{LLM}((\text{questions}))$, $C.\text{sup.}.\text{src}$ denotes the set that may appear in the image feature, and $C.\text{sup.}.\text{tgt}$ denotes the set of the desired substitutes.

[0042] Next, command module 230 may find the best match of the patch feature via search, such that:

$$[00008] T_{F^{(j)}} = \underset{k \in [C^{\text{src}}]}{\text{argmax}} \quad g(F^{(j)}, T_k), \quad \text{Equation(7)}$$

where $g(\cdot, \cdot)$ denotes a similarity measure (e.g., dot product). In some embodiments, the search may be further improved by taking advantage of advanced optimization techniques such as text inversion, prompt tuning, etc.

[0043] After the search is complete, command module 230 may then manipulate the dense feature descriptor F' by replacing $F'.\text{sup.}(j)$ with textual features

$$[00009] h(T_{F^{(j)}}, \{T_K\}_{k \in [C^{\text{tgt}}]})$$

under conditions such as similarity above a certain threshold or stochasticity. The function h may be a human prior or LLMs that conceptually answer the question of what may be a plausible substitute from $C.\text{sup.}.\text{tgt}$ under the current context.

[0044] To generate cross-modality features, for example, command module 230 may calculate textual features for a predefined set of natural language concepts (e.g., road, car), identify the best-matching textual feature for each image pixel, along with the degree of similarity, and replace the image feature at pixels where the similarity exceeds a pre-determined threshold with the corresponding textual feature. In some embodiments, the pre-determined threshold may be adjusted such that a value (e.g., 0, null) leads to driving based solely on textual features, while a higher value (e.g., 1, inf) leads to driving based exclusively on image features. In some embodiments, the selection of pixels for generating cross-modality features may be specified by a pre-determined function (e.g., a random function), such that only a portion of the image as given by the pre-determined function may have cross-modality features.

[0045] In some embodiments, command module 230 may consult LLMs to generate a base set of natural language concepts relevant to a driving scenario. For example, command module 230 may

request a set of terms relating to “any non-drivable objects or entities likely to appear in a rural driving scenario”. In some embodiments, the driving scenarios may be pre-determined. In some embodiments, a user may select and remove terms from the set of terms generated by an LLM. [0046] FIG. 6 illustrates a flowchart of a method 600 that is associated with strategies for using multi-modal foundation models. Method 600 will be discussed from the perspective of the E2E driving system 170 of FIGS. 1 and 2. While method 600 is discussed in combination with the E2E driving system 170, it should be appreciated that the method 600 is not limited to being implemented within E2E driving system 170 but is instead one example of a system that may implement method 600.

[0047] At step 610, command module 230 may receive images and a foundation multi-model. For example, command module 230 may receive images from sensor data 250 and a foundation multi-modal, such as CLIP, DINO, or BLIP via prediction model 260.

[0048] At step 620, command module 230 may select a mask set. For example, command module 230 may select a mask set according to Equation (4) and Equation (5).

[0049] At step 630, command module 230 may modify the foundation multi-model to include query, key, and value matrices.

[0050] At step 640, command module 230 may apply the mask set to the foundation multi-model to obtain patch-aligned features.

[0051] FIG. 1 will now be discussed in full detail as an example environment within which the system and methods disclosed herein may operate. In some instances, vehicle 100 is configured to switch selectively between various modes, such as an autonomous mode, one or more semi-autonomous operational modes, a manual mode, etc. Such switching may be implemented in a suitable manner, now known, or later developed. “Manual mode” means that all of or a majority of the navigation/maneuvering of the vehicle is performed according to inputs received from a user (e.g., human driver). In one or more arrangements, vehicle 100 may be a conventional vehicle that is configured to operate in only a manual mode.

[0052] In one or more embodiments, vehicle 100 is an autonomous vehicle. As used herein, “autonomous vehicle” refers to a vehicle that operates in an autonomous mode. “Autonomous mode” refers to using one or more computing systems to control vehicle 100, such as providing navigation/maneuvering of vehicle 100 along a travel route, with minimal or no input from a human driver. In one or more embodiments, vehicle 100 is either highly automated or completely automated. In one embodiment, vehicle 100 is configured with one or more semi-autonomous operational modes in which one or more computing systems perform a portion of the navigation/maneuvering of the vehicle along a travel route, and a vehicle operator (i.e., driver) provides inputs to the vehicle to perform a portion of the navigation/maneuvering of vehicle 100 along a travel route.

[0053] Vehicle 100 may include one or more processors 110. In one or more arrangements, processor(s) 110 may be a main processor of vehicle 100. For instance, processor(s) 110 may be an electronic control unit (ECU). Vehicle 100 may include one or more data stores 115 for storing one or more types of data. Data store(s) 115 may include volatile memory, non-volatile memory, or both. Examples of suitable data store(s) 115 include RAM (Random Access Memory), flash memory, ROM (Read Only Memory), PROM (Programmable Read-Only Memory), EPROM (Erasable Programmable Read-Only Memory), EEPROM (Electrically Erasable Programmable Read-Only Memory), registers, magnetic disks, optical disks, hard drives, or any other suitable storage medium, or any combination thereof. Data store(s) 115 may be a component of processor(s) 110, or data store 115 may be operatively connected to processor(s) 110 for use thereby. The term “operatively connected,” as used throughout this description, may include direct or indirect connections, including connections without direct physical contact.

[0054] In one or more arrangements, data store(s) 115 may include map data 116. Map data 116 may include maps of one or more geographic areas. In some instances, map data 116 may include

information or data on roads, traffic control devices, road markings, structures, features, landmarks, or any combination thereof in the one or more geographic areas. Map data **116** may be in any suitable form. In some instances, map data **116** may include aerial views of an area. In some instances, map data **116** may include ground views of an area, including 360-degree ground views. Map data **116** may include measurements, dimensions, distances, information, or any combination thereof for one or more items included in map data **116**. Map data **116** may also include measurements, dimensions, distances, information, or any combination thereof relative to other items included in map data **116**. Map data **116** may include a digital map with information about road geometry. Map data **116** may be high quality, highly detailed, or both.

[0055] In one or more arrangements, map data **116** may include one or more terrain maps **117**. Terrain map(s) **117** may include information about the ground, terrain, roads, surfaces, other features, or any combination thereof of one or more geographic areas. Terrain map(s) **117** may include elevation data in the one or more geographic areas. Terrain map(s) **117** may be high quality, highly detailed, or both. Terrain map(s) **117** may define one or more ground surfaces, which may include paved roads, unpaved roads, land, and other things that define a ground surface.

[0056] In one or more arrangements, map data **116** may include one or more static obstacle maps **118**. Static obstacle map(s) **118** may include information about one or more static obstacles located within one or more geographic areas. A “static obstacle” is a physical object whose position does not change or substantially change over a period of time and whose size does not change or substantially change over a period of time. Examples of static obstacles include trees, buildings, curbs, fences, railings, medians, utility poles, statues, monuments, signs, benches, furniture, mailboxes, large rocks, hills. The static obstacles may be objects that extend above ground level. The one or more static obstacles included in static obstacle map(s) **118** may have location data, size data, dimension data, material data, other data, or any combination thereof, associated with it. Static obstacle map(s) **118** may include measurements, dimensions, distances, information, or any combination thereof for one or more static obstacles. Static obstacle map(s) **118** may be high quality, highly detailed, or both. Static obstacle map(s) **118** may be updated to reflect changes within a mapped area.

[0057] Data store(s) **115** may include sensor data **119**. In this context, “sensor data” means any information about the sensors that vehicle **100** is equipped with, including the capabilities and other information about such sensors. As will be explained below, vehicle **100** may include sensor system **120**. Sensor data **119** may relate to one or more sensors of sensor system **120**. As an example, in one or more arrangements, sensor data **119** may include information on one or more LIDAR sensors **124** of sensor system **120**.

[0058] In some instances, at least a portion of map data **116** or sensor data **119** may be located in data stores(s) **115** located onboard vehicle **100**. Alternatively, or in addition, at least a portion of map data **116** or sensor data **119** may be located in data stores(s) **115** that are located remotely from vehicle **100**.

[0059] As noted above, vehicle **100** may include sensor system **120**. Sensor system **120** may include one or more sensors. “Sensor” means any device, component, or system that may detect or sense something. The one or more sensors may be configured to sense, detect, or perform both in real-time. As used herein, the term “real-time” means a level of processing responsiveness that a user or system senses as sufficiently immediate for a particular process or determination to be made, or that enables the processor to keep up with some external process.

[0060] In arrangements in which sensor system **120** includes a plurality of sensors, the sensors may work independently from each other. Alternatively, two or more of the sensors may work in combination with each other. In such an embodiment, the two or more sensors may form a sensor network. Sensor system **120**, the one or more sensors, or both may be operatively connected to processor(s) **110**, data store(s) **115**, another element of vehicle **100** (including any of the elements shown in FIG. **1**), or any combination thereof. Sensor system **120** may acquire data of at least a

portion of the external environment of vehicle **100** (e.g., nearby vehicles).

[0061] Sensor system **120** may include any suitable type of sensor. Various examples of different types of sensors will be described herein. However, it will be understood that the embodiments are not limited to the particular sensors described. Sensor system **120** may include one or more vehicle sensors **121**. Vehicle sensor(s) **121** may detect, determine, sense, or acquire in a combination thereof information about vehicle **100** itself. In one or more arrangements, vehicle sensor(s) **121** may be configured to detect, sense, or acquire in a combination thereof position and orientation changes of vehicle **100**, such as, for example, based on inertial acceleration. In one or more arrangements, vehicle sensor(s) **121** may include one or more accelerometers, one or more gyroscopes, an inertial measurement unit (IMU), a dead-reckoning system, a global navigation satellite system (GNSS), a global positioning system (GPS), a navigation system **147**, other suitable sensors, or any combination thereof. Vehicle sensor(s) **121** may be configured to detect, sense, or acquire in a combination thereof one or more characteristics of vehicle **100**. In one or more arrangements, vehicle sensor(s) **121** may include a speedometer to determine a current speed of vehicle **100**.

[0062] Alternatively, or in addition, sensor system **120** may include one or more environment sensors **122** configured to acquire, sense, or acquire in a combination thereof driving environment data. “Driving environment data” includes data or information about the external environment in which an autonomous vehicle is located or one or more portions thereof. For example, environment sensor(s) **122** may be configured to detect, quantify, sense, or acquire in any combination thereof obstacles in at least a portion of the external environment of vehicle **100**, information/data about such obstacles, or a combination thereof. Such obstacles may be comprised of stationary objects, dynamic objects, or a combination thereof. Environment sensor(s) **122** may be configured to detect, measure, quantify, sense, or acquire in any combination thereof other things in the external environment of vehicle **100**, such as, for example, lane markers, signs, traffic lights, traffic signs, lane lines, crosswalks, curbs proximate to vehicle **100**, off-road objects, etc.

[0063] Various examples of sensors of sensor system **120** will be described herein. The example sensors may be part of the one or more environment sensor(s) **122**, the one or more vehicle sensors **121**, or both. However, it will be understood that the embodiments are not limited to the particular sensors described.

[0064] As an example, in one or more arrangements, sensor system **120** may include one or more radar sensors **123**, one or more LIDAR sensors **124**, one or more sonar sensors **125**, one or more cameras **126**, or any combination thereof. In one or more arrangements, camera(s) **126** may be high dynamic range (HDR) cameras or infrared (IR) cameras.

[0065] Vehicle **100** may include an input system **130**. An “input system” includes any device, component, system, element or arrangement or groups thereof that enable information/data to be entered into a machine. Input system **130** may receive an input from a vehicle passenger (e.g., a driver or a passenger). Vehicle **100** may include an output system **135**. An “output system” includes any device, component, or arrangement or groups thereof that enable information/data to be presented to a vehicle passenger (e.g., a person, a vehicle passenger, etc.).

[0066] Vehicle **100** may include one or more vehicle systems **140**. Various examples of vehicle system(s) **140** are shown in FIG. **1**. However, vehicle **100** may include more, fewer, or different vehicle systems. It should be appreciated that although particular vehicle systems are separately defined, each or any of the systems or portions thereof may be otherwise combined or segregated via hardware, software, or a combination thereof within vehicle **100**. Vehicle **100** may include a propulsion system **141**, a braking system **142**, a steering system **143**, throttle system **144**, a transmission system **145**, a signaling system **146**, a navigation system **147**, other systems, or any combination thereof. Each of these systems may include one or more devices, components, or combinations thereof, now known or later developed.

[0067] Navigation system **147** may include one or more devices, applications, or combinations

thereof, now known or later developed, configured to determine the geographic location of the vehicle **100**, to determine a travel route for vehicle **100**, or to determine both. Navigation system **147** may include one or more mapping applications to determine a travel route for vehicle **100**. Navigation system **147** may include a global positioning system, a local positioning system, a geolocation system, or any combination thereof.

[0068] Processor(s) **110**, E2E driving system **170**, automated driving module(s) **160**, or any combination thereof may be operatively connected to communicate with various aspects of vehicle system(s) **140** or individual components thereof. For example, returning to FIG. **1**, processor(s) **110**, automated driving module(s) **160**, or a combination thereof may be in communication to send or receive information from various aspects of vehicle system(s) **140** to control the movement, speed, maneuvering, heading, direction, etc. of vehicle **100**. Processor(s) **110**, E2E driving system **170**, automated driving module(s) **160**, or any combination thereof may control some or all of these vehicle system(s) **140** and, thus, may be partially or fully autonomous.

[0069] Processor(s) **110**, E2E driving system **170**, automated driving module(s) **160**, or any combination thereof may be operable to control at least one of the navigation or maneuvering of vehicle **100** by controlling one or more of vehicle systems **140** or components thereof. For instance, when operating in an autonomous mode, processor(s) **110**, E2E driving system **170**, automated driving module(s) **160**, or any combination thereof may control the direction, speed, or both of vehicle **100**. Processor(s) **110**, E2E driving system **170**, automated driving module(s) **160**, or any combination thereof may cause vehicle **100** to accelerate (e.g., by increasing the supply of fuel provided to the engine), decelerate (e.g., by decreasing the supply of fuel to the engine, by applying brakes), change direction (e.g., by turning the front two wheels), or perform any combination thereof. As used herein, “cause” or “causing” means to make, force, compel, direct, command, instruct, enable, or in any combination thereof an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner.

[0070] Vehicle **100** may include one or more actuators **150**. Actuator(s) **150** may be any element or combination of elements operable to modify, adjust, alter, or in any combination thereof one or more of vehicle systems **140** or components thereof to responsive to receiving signals or other inputs from processor(s) **110**, automated driving module(s) **160**, or a combination thereof. Any suitable actuator may be used. For instance, actuator(s) **150** may include motors, pneumatic actuators, hydraulic pistons, relays, solenoids, and piezoelectric actuators, just to name a few possibilities.

[0071] Vehicle **100** may include one or more modules, at least some of which are described herein. The modules may be implemented as computer-readable program code that, when executed by processor(s) **110**, implement one or more of the various processes described herein. One or more of the modules may be a component of processor(s) **110**, or one or more of the modules may be executed on or distributed among other processing systems to which processor(s) **110** is operatively connected. The modules may include instructions (e.g., program logic) executable by processor(s) **110**. Alternatively, or in addition, data store(s) **115** may contain such instructions.

[0072] In one or more arrangements, one or more of the modules described herein may include artificial or computational intelligence elements, e.g., neural network, fuzzy logic, or other machine learning algorithms. Further, in one or more arrangements, one or more of the modules may be distributed among a plurality of the modules described herein. In one or more arrangements, two or more of the modules described herein may be combined into a single module.

[0073] Vehicle **100** may include one or more autonomous driving modules **160**. Automated driving module(s) **160** may be configured to receive data from sensor system **120** or any other type of system capable of capturing information relating to vehicle **100**, the external environment of the vehicle **100**, or a combination thereof. In one or more arrangements, automated driving module(s) **160** may use such data to generate one or more driving scene models. Automated driving module(s) **160** may determine position and velocity of vehicle **100**. Automated driving module(s) **160** may

determine the location of obstacles, obstacles, or other environmental features including traffic signs, trees, shrubs, neighboring vehicles, pedestrians, etc.

[0074] Automated driving module(s) **160** may be configured to receive, determine, or in a combination thereof location information for obstacles within the external environment of vehicle **100**, which may be used by processor(s) **110**, one or more of the modules described herein, or any combination thereof to estimate: a position or orientation of vehicle **100**; a vehicle position or orientation in global coordinates based on signals from a plurality of satellites or other geolocation systems; or any other data/signals that could be used to determine a position or orientation of vehicle **100** with respect to its environment for use in either creating a map or determining the position of vehicle **100** in respect to map data.

[0075] Automated driving module(s) **160** either independently or in combination with E2E driving system **170** may be configured to determine travel path(s), current autonomous driving maneuvers for vehicle **100**, future autonomous driving maneuvers, modifications to current autonomous driving maneuvers, etc. Such determinations by automated driving module(s) **160** may be based on data acquired by sensor system **120**, driving scene models, data from any other suitable source such as determinations from sensor data **250**, or any combination thereof. In general, automated driving module(s) **160** may function to implement different levels of automation, including advanced driving assistance (ADAS) functions, semi-autonomous functions, and fully autonomous functions. “Driving maneuver” means one or more actions that affect the movement of a vehicle. Examples of driving maneuvers include accelerating, decelerating, braking, turning, moving in a lateral direction of vehicle **100**, changing travel lanes, merging into a travel lane, and reversing, just to name a few possibilities. Automated driving module(s) **160** may be configured to implement driving maneuvers. Automated driving module(s) **160** may cause, directly or indirectly, such autonomous driving maneuvers to be implemented. As used herein, “cause” or “causing” means to make, command, instruct, enable, or in any combination thereof an event or action to occur or at least be in a state where such event or action may occur, either in a direct or indirect manner. Automated driving module(s) **160** may be configured to execute various vehicle functions, whether individually or in combination, to transmit data to, receive data from, interact with, or to control vehicle **100** or one or more systems thereof (e.g., one or more of vehicle systems **140**).

[0076] Detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are intended only as examples. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the aspects herein in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of possible implementations. Various embodiments are shown in FIGS. **1-6**, but the embodiments are not limited to the illustrated structure or application.

[0077] The flowcharts and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowcharts or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.

[0078] The systems, components, or processes described above may be realized in hardware or a combination of hardware and software and may be realized in a centralized fashion in one processing system or in a distributed fashion where different elements are spread across several interconnected processing systems. Any kind of processing system or another apparatus adapted for

carrying out the methods described herein is suited. A typical combination of hardware and software may be a processing system with computer-usable program code that, when being loaded and executed, controls the processing system such that it carries out the methods described herein. The systems, components, or processes also may be embedded in a computer-readable storage, such as a computer program product or other data programs storage device, readable by a machine, tangibly embodying a program of instructions executable by the machine to perform methods and processes described herein. These elements also may be embedded in an application product which comprises all the features enabling the implementation of the methods described herein and, which when loaded in a processing system, is able to carry out these methods.

[0079] Furthermore, arrangements described herein may take the form of a computer program product embodied in one or more computer-readable media having computer-readable program code embodied, e.g., stored, thereon. Any combination of one or more computer-readable media may be utilized. The computer-readable medium may be a computer-readable signal medium or a computer-readable storage medium. The phrase “computer-readable storage medium” means a non-transitory storage medium. A computer-readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer-readable storage medium would include the following: a portable computer diskette, a hard disk drive (HDD), a solid-state drive (SSD), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), a digital versatile disc (DVD), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer-readable storage medium may be any tangible medium that may contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0080] Generally, modules as used herein include routines, programs, objects, components, data structures, and so on that perform particular tasks or implement particular data types. In further aspects, a memory generally stores the noted modules. The memory associated with a module may be a buffer or cache embedded within a processor, a RAM, a ROM, a flash memory, or another suitable electronic storage medium. In still further aspects, a module as envisioned by the present disclosure is implemented as an application-specific integrated circuit (ASIC), a hardware component of a system on a chip (SoC), as a programmable logic array (PLA), or as another suitable hardware component that is embedded with a defined configuration set (e.g., instructions) for performing the disclosed functions.

[0081] Program code embodied on a computer-readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber, cable, RF, etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the present arrangements may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java™, Smalltalk, C++, or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on a user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer, or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0082] The terms “a” and “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and “having,” as used herein, are defined as comprising (i.e., open language). The phrase “at least one of . . . and . . .” as used herein

refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase “at least one of A, B, and C” includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC, or ABC).

[0083] Aspects herein may be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

Claims

1. A system, comprising: a processor; and a memory communicably coupled to the processor and storing machine-readable instructions that, when executed by the processor, cause the processor to: receive images and a foundation multi-model; select a mask set; modify the foundation multi-model to include query, key, and value matrices; and apply the mask set to the foundation multi-model to obtain patch-aligned features.
2. The system of claim 1, wherein the foundation multi-model is based on CLIP, DINO, or BLIP.
3. The system of claim 1, wherein the mask set is determined based on a distance function between a first patch and a second patch.
4. The system of claim 1, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: obtain a set of concepts in natural language and computing their corresponding textual features.
5. The system of claim 4, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: search the patch-aligned features to obtain a match with each textual feature.
6. The system of claim 5, wherein the machine-readable instructions that, when executed by the processor, further includes causing the processor to: replace at least one patch-aligned feature with at least one textual feature.
7. The system of claim 5, wherein the match is based on a function determining an estimate of similarity and whether the function returns a result above a pre-determined threshold.
8. A non-transitory computer-readable medium including instructions that when executed by one or more processors cause the one or more processors to: receive images and a foundation multi-model; select a mask set; modify the foundation multi-model to include query, key, and value matrices; and apply the mask set to the foundation multi-model to obtain patch-aligned features.
9. The non-transitory computer-readable medium of claim 8, wherein the foundation multi-model is based on CLIP, DINO, or BLIP.
10. The non-transitory computer-readable medium of claim 8, wherein the mask set is determined based on a distance function between a first patch and a second patch.
11. The non-transitory computer-readable medium of claim 8, wherein the instructions further include to: obtain a set of concepts in natural language and computing their corresponding textual features.
12. The non-transitory computer-readable medium of claim 11, wherein the instructions further include to: search the patch-aligned features to obtain a match with each textual feature.
13. The non-transitory computer-readable medium of claim 12, wherein the instructions further include to: replace at least one patch-aligned feature with at least one textual feature.
14. A method, comprising: receiving images and a foundation multi-model; selecting a mask set; modifying the foundation multi-model to include query, key, and value matrices; and applying the mask set to the foundation multi-model to obtain patch-aligned features.
15. The method of claim 14, wherein the foundation multi-model is based on CLIP, DINO, or BLIP.
16. The method of claim 14, wherein the mask set is determined based on a distance function between a first patch and a second patch.

- 17.** The method of claim 14, further comprising: obtaining a set of concepts in natural language and computing their corresponding textual features.
- 18.** The method of claim 17, further comprising: searching the patch-aligned features to obtain a match with each textual feature.
- 19.** The method of claim 18, further comprising: replacing at least one patch-aligned feature with at least one textual feature.
- 20.** The method of claim 18, wherein the match is based on a function determining an estimate of similarity and whether the function returns a result above a pre-determined threshold.
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